

MODULE 2.6

Cybersecurity Resilience

Network Defense | Cisco Networking Academy

Prepared by: Kudzaishe Majeza

Cisco Networking Academy | 2026

Topics Covered:

2.6.2 • High Availability

2.6.3 • The Five Nines

2.6.4 • Single Points of Failure

2.6.6 / 2.6.7 • N+1 Redundancy & RAID

2.6.8–2.6.10 • STP, Router & Location Redundancy

2.6.11–2.6.14 • Resilient Design, Backups & Power

2.6.15 / 2.6.17 • HVAC & Physical Facility Threats

2.6.2 High Availability

Designing systems to avoid downtime and ensure continuous availability

What is High Availability?

- High availability describes systems designed to avoid downtime as much as possible.
- The continuous availability of information systems is imperative to organisations and modern life — we all rely on these systems more than ever before.
- For @Apollo, the goal is to minimise downtime of mission-critical processes.

Three Design Principles of High Availability

- Eliminate Single Points of Failure — Identify every component whose failure would bring down the entire system and replace or duplicate it.
- Provide for Reliable Crossover — Ensure that when a component fails, the standby system takes over seamlessly without interruption.
- Detect Failures as They Occur — Continuously monitor systems so that failures are identified and acted on immediately, minimising impact.

2.6.3 The Five Nines

Achieving 99.999% availability — the gold standard for uptime

What Are the Five Nines?

- Every organisation wants to operate uninterrupted, even under extreme conditions like an attack.
- 'Five Nines' refers to a target availability rate of 99.999%.
- In practice, this means less than 5.26 minutes of downtime per year.
- Achieving five nines requires redundancy, failover systems and resilient architecture.

Availability Comparison

- 99% → 3.65 days of downtime per year
 - 99.9% → 8.77 hours of downtime per year
 - 99.99% → 52.6 minutes of downtime per year
 - 99.999% → 5.26 minutes of downtime per year ✓ (Five Nines)
-
- Common steps to reach five nines: redundant hardware, failover systems, load balancing, proactive monitoring and fast recovery procedures.

2.6.4 Single Points of Failure

Identifying and eliminating weak links in the system chain

What is a Single Point of Failure (SPOF)?

- A single point of failure is any component whose failure causes the entire system to fail.
- It can be a piece of hardware, a process, a piece of data, or an essential utility.
- For @Apollo, an SPOF could be a single router, a single power supply, or a single internet connection.

How to Eliminate SPOFs

- Modify critical operations so they do not rely on any single element.
- Use hot stand-by devices — backup components that can take over immediately on failure.
- Add redundant connections or pathways so traffic can re-route if one path fails.
- Build redundant components into every critical operation so no single element is relied upon entirely.

2.6.6 / 2.6.7 N+1 Redundancy & RAID

Ensuring availability through backup components and redundant storage

N+1 Redundancy

- N+1 redundancy means every component (N) has at least one backup (+1) ready to take over.
- Example: A car has 4 tyres (N) and a spare tyre (+1) in the boot.
- In a data center: servers, power supplies, switches and routers all have standby units ready to come online if the primary fails.
- Note: N+1 is not a fully redundant system — it only covers a single failure at a time.

RAID — Redundant Array of Independent Disks

- RAID spreads data across multiple drives to protect against a single disk failure and improves data retrieval speed.
- Mirroring — Duplicates all data onto a second drive.
- Striping — Writes data across multiple drives in consecutive segments for speed.
- Parity — Striping with checksums stored on a third drive for error recovery.
- Most RAID levels combine these approaches to balance speed, redundancy and storage.

2.6.8 Spanning Tree Protocol (STP)

Preventing network loops while maintaining redundant physical paths

The Problem with Redundant Links

- Redundant physical links between switches eliminate single points of failure.
- But multiple paths create loops — frames circulate endlessly, consuming all bandwidth and bringing the network down.

How STP Solves It

- STP prevents loops by intentionally blocking redundant paths, keeping only one logical path active between any two destinations.
- The physical paths still exist — STP just disables them to prevent loops.
- When the active link fails, STP recalculates and automatically unblocks the standby path so traffic continues.
- STP stages: Blocking → Listening → Learning → Forwarding.
- The result: redundancy is maintained without the risk of loops.

2.6.9 / 2.6.10 Router & Location Redundancy

Eliminating the default gateway as a single point of failure

Router Redundancy

- The default gateway router is a common SPOF — if it fails, all devices lose network access.
- A standby router is installed alongside the active router.
- Both share a virtual IP address used as the default gateway by all end devices.
- A redundancy protocol determines which router is active at any time. On failure, the standby takes over automatically.

Location Redundancy

- Organisations may replicate systems across multiple physical sites to survive a site-level failure.
- Synchronous Replication — Both sites updated in real time. High bandwidth required. Sites must be close to minimise latency.
- Asynchronous Replication — Near-real-time sync. Lower bandwidth requirements. Sites can be further apart.
- Point-in-Time Replication — Backup updated at set intervals. Most bandwidth-efficient. Best for lower-priority data.

2.6.11 / 2.6.12 Resilient Design & Data Backups

Building tolerant systems and protecting data through regular backups

Resilient Design

- Resiliency refers to methods that make a system or network tolerant of failure.
- Example: Redundant links with STP provide an alternate path if a link fails.
- Routing protocols also provide resiliency — administrators can fine-tune failover settings to reduce disruption for users.
- Resilient design starts with understanding business needs, then designing to protect them.

System and Data Backups

- Data can be lost through cyberattack, hardware failure, disaster or human error.
- Frequency — Full backup monthly or weekly; partial backups of changed data more frequently.
- Storage — Rotate backups to an approved off-site location on a regular schedule.
- Security — Protect backup media with strong passwords.
- Validation — Always verify backups — a backup that cannot be restored is useless.

2.6.14 / 2.6.15 Power & HVAC Systems

Managing electrical power and environmental controls in critical facilities

Electrical Power Systems

- A continuous power supply is essential for data centers and server facilities.
- Data center power should be on a separate supply from the rest of the building.
- Use redundant feeds from two or more electrical substations.
- Implement power conditioning and UPS (Uninterruptible Power Supply) to allow graceful shutdown during outages.
- Power events: Spike (momentary high voltage), Surge (prolonged high voltage), Sag/Brownout (low voltage), Blackout (complete loss).

HVAC Systems

- HVAC controls temperature, humidity and airflow — all critical for IT hardware health.
- All hardware has environmental requirements — operating outside these ranges causes failures and shortens equipment lifespan.
- Modern HVAC systems connect to the internet for remote monitoring, introducing significant security vulnerabilities.
- HVAC contractors and third-party vendors often have broad access to these systems — apply the same access controls as to any other critical infrastructure.

2.6.17 Managing Threats to Physical Facilities

Countermeasures for physical domain cybersecurity threats at @Apollo

Why Physical Security Matters

- Physical facilities face threats including unauthorized access, theft, social engineering, power interruptions and environmental hazards.
- No digital control can compensate for an attacker who gains physical access to systems.

Countermeasures

- Access control and CCTV coverage at all entrances
- Badge encryption for secure entry access
- Guest visitor policies and escort procedures
- Physical penetration testing of the building
- Asset tagging and inventory management
- Disaster recovery planning and business continuity planning
- Regular security awareness training for all staff
- Redundant power, UPS and environmental control systems

2.6 Summary — Cybersecurity Resilience

Key concepts from this module

Availability & Redundancy

- High Availability — design to eliminate downtime
- Five Nines — 99.999% uptime, <5.26 min downtime / year
- SPOF — identify and eliminate every single point of failure
- N+1 Redundancy — at least one backup for every component
- RAID — mirroring, striping and parity across multiple disks
- STP — prevents loops while keeping redundant paths available
- Router Redundancy — standby router with virtual IP failover
- Location Redundancy — synchronous / asynchronous / point-in-time replication

Infrastructure Protection

- Resilient Design — build failure tolerance based on business needs
- Data Backups — frequency, off-site storage, password protection, validation
- Power — redundant feeds, UPS, power conditioning, separate supply
- HVAC — temperature and humidity control; secure third-party access
- Physical Threats — access control, CCTV, badges, guest policies, security testing, disaster recovery
- Key Principle: Resilience is not just redundancy — understand what the business needs and design to protect it.